

⇒ Dijkstra's Algorithm

```
#include <stdio.h>
```

```
#define MAX 10
```

```
#define INF 99999
```

```
void Dijkstra(int n, int G[MAX][MAX], int src) {
```

```
    int dist[MAX], visited[MAX];
```

```
    int i, j, u, min;
```

```
    for (i = 0; i < n; i++) {
```

```
        dist[i] = INF;
```

```
        visited[i] = 0;
```

```
    }
```

```
    dist[src] = 0;
```

```
    for (i = 0; i < n - 1; i++) {
```

```
        min = INF;
```

```
    }
```

```
    for (j = 0; j < n; j++) {
```

```
        if (!visited[j] && dist[j] <= min) {
```

```
            min = dist[j];
```

```
            u = j;
```

```
        }
```

```
    }
```

```

visited[u] = 1;
for (j=0; j<n; j++) {
    if (!visited[j] && G[u][j] != INF &&
        dist[u] + G[u][j] < dist[j]) {
        dist[j] = dist[u] + G[u][j];
    }
}

printf("\n Vertex \n Distance from source \n");
for (i=0; i<n; i++) {
    if (dist[i] == INF)
        printf("%d\t INF\t ", i);
    else
        printf("%d\t %d\t ", i, dist[i]);
}
}

```

```

int main() {
    int n, i, j, src;
    int G[MAX][MAX];
    printf("Enter no. of vertices: ");
    scanf("%d", &n);
    printf("Enter the weight matrix: \n");
    for (i=0; i<n; i++) {
        for (j=0; j<n; j++) {
            scanf("%d", &G[i][j]);
        }
    }
}

```



```
printf("Enter source vertex: ");
scanf("%d", &src);
dijkstra(m, n, src);
return 0;
}
```

O/P: Enter no. of vertices: 4

Enter weight matrix:

2 10 5 7
6 0 7 1
0 6 0 4
6 3 9 0

Enter source vertex: 1

vertex	Distance from source
0	6
1	0
2	7
3	1

[Handwritten signature]